Hugo Marquerie 18/03/2025

Probabilidad total

Teorema 1 (Probabilidad total). *Sea* $(X, \Sigma, \mathbb{P})$ *un espacio de probabilidad y* $\{A_i\}_{i\in\mathbb{N}} \subset \Sigma$ *una partición de* Ω *tal que* $\forall i \in \mathbb{N} : \mathbb{P}(A_i) \neq 0$

$$\implies \forall B \in \Sigma : \mathbb{P}(B) = \sum_{i\in\mathbb{N}} \mathbb{P}(A_i) \cdot \mathbb{P}(B \mid A_i).$$

***Demostración*:** Sea $B \in \Sigma$, entonces $B = \bigsqcup_{i\in\mathbb{N}} B \cap A_i$, y por la `probabilidad-condicionada/`Definició

$$\mathbb{P}(B) \overset{\overset{\text{aditividad}}{\downarrow}}{=} \sum_{i\in\mathbb{N}} \mathbb{P}(B \cap A_i) \overset{\overset{\text{def}}{\downarrow}}{=} \sum_{i\in\mathbb{N}} \mathbb{P}(A_i) \cdot \mathbb{P}(B \mid A_i).$$

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